

Zhexi Lu

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Education

Ph.D. in Computer Science
Rensselaer Polytechnic Institute

2024 – Expected May 2029
Troy, NY, USA

- Advisor: Prof. Lei Yu

B.S. in Information Security
Northeastern University (China)

2020 – 2024
Shenyang, China

- GPA: 4.134/5.0 (Top 7%)

Publications

1. [C1] **In-Context Probing for Membership Inference in Fine-Tuned Language Models**
Zhexi Lu, Hongliang Chi, Nathalie Baracaldo, Swanand Ravindra Kadhe, Yuseok Jeon, Lei Yu
Network and Distributed System Security (NDSS) Symposium (Acceptance Rate: 17.9%) 2026
2. [C2] **Activation-Conditioned Self-Distillation**
Zhexi Lu, Subhajit Chaudhury, Tejaswini Pedapati, Keerthiram Murugesan, Lei Yu
Preprint
3. [C3] **Leaky Students: Membership Inference against On-Policy Distillation**
Zhexi Lu, Mingzhi Zhu, Stacy Patterson, Lei Yu
Preprint
4. [C4] **Internalizing State Tracking for Reliable Multi-Turn Tool Agents**
Zhexi Lu, Subhajit Chaudhury, Tejaswini Pedapati, Keerthiram Murugesan, Lei Yu
Preprint
5. [C5] **FDRShare: A Fully Decentralized and Redactable EHRs Sharing Scheme with Constant-Size Ciphertexts**
Zhichao Li*, Zhexi Lu*, Lingshuai Wang, Qiang Wang, Che Bian (*Equal contribution)
International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP) 2023

Work and Research Experience

Summer Visiting Student
IBM Research

May 2026 – Aug. 2026
Yorktown Heights, NY

- Topic: On-policy self-distillation for LLM reasoning.
- Proposed **Activation-Conditioned Self-Distillation (ACSD)**, which steers a frozen copy of the model with an activation contrast between its own correct-within-budget and remaining trajectories, yielding a teacher without reference solutions.
- Achieved the best mean accuracy on 4 math benchmarks for all 5 LLMs tested, over SFT, GRPO, OPSD, and RLCSO; on DeepSeek-R1-0528-Qwen3-8B, raised math accuracy 67.6%→71.9% and LiveCodeBench v6 pass@12 65.7%→70.9%.
- Showed that ACSD sustains token-level supervision on long responses, and that its directions transfer across datasets and can be extracted from as few as 16 problems. [*Preprint*]

Summer Visiting Student
IBM Research

May 2025 – Aug. 2025
Yorktown Heights, NY

- Topic: Failure detection in multi-turn agentic interactions.
- Designed a fine-grained failure taxonomy for agentic systems and built an annotation pipeline to collect failure traces from GPT-4o, Claude-3.7, and xLAM on the BFCL multi-turn dataset.
- Created **FailureCall-Bench**, a manually annotated benchmark of 510 dialogues spanning five error categories.
- Evaluated 7 LLMs on detection, first-error identification, and classification tasks; accuracy ranged 28.6–41.4%, exposing a gap between error detection and causal diagnosis.

Research Assistant
Rensselaer Polytechnic Institute

Sep. 2024 – Present
Troy, NY

- Topic: Membership inference attacks against fine-tuned LLMs.
- Identified the *Optimization Gap*—the disparity in remaining loss-reduction potential—as a fundamental membership signal; non-members exhibit $2.9\times$ larger loss reduction than members in controlled experiments.
- Designed **In-Context Probing (ICP)**, a training-free black-box attack that approximates the Optimization Gap via two probing strategies: semantic-similarity retrieval and self-perturbation.
- Achieved state-of-the-art results over 7 baselines on 3 datasets and 3 LLMs (AUC 0.942 on HealthcareMagic; TPR@1%FPR 0.518 on CNN-DM, $2.6\times$ over ReCaLL). [*NDSS 2026*]

Technical Skills

Languages

Python, C/C++, $\text{\LaTeX}$

ML/NLP

PyTorch, Hugging Face Transformers, vLLM

Tools

Git, Linux, Docker, Slurm, Weights & Biases

Honors and Awards

- **Google PhD Fellowship Nominee**, nominated by RPI (up to 4 nominees per university) 2026
- **Second Prize**, National College Student Information Security Competition 2023
- **Third Prize**, Mathematical Contest in Modeling (MCM/ICM) 2023
- **Third Prize**, Contemporary Undergraduate Mathematical Contest in Modeling 2022
- **Outstanding Student Leader Award**, Northeastern University (China) 2023
- **Northeastern University Scholarship** 2020 – 2023

Leadership & Service

- **Class President**, School of Computer Science, Northeastern University (China) 2020 – 2024
- **Reviewer:** NeurIPS 2025, AsianCCS 2025
- **Artifact Reviewer:** ACM CCS 2025

Relevant Courses

Computer Science

Algorithms, Deep Learning, Natural Language Processing, Machine Learning, Operating Systems, Programming Languages, Trustworthy AI

Mathematics

Linear Algebra, Probability, Cryptography